Lecture 6: Atmospheres II
Clouds, Weather, & Climate

Planetary Systems

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Jupiter’s atmosphere. Credit: NASA/JPL-Caltech/SwRI/MSSS

Learning objectives

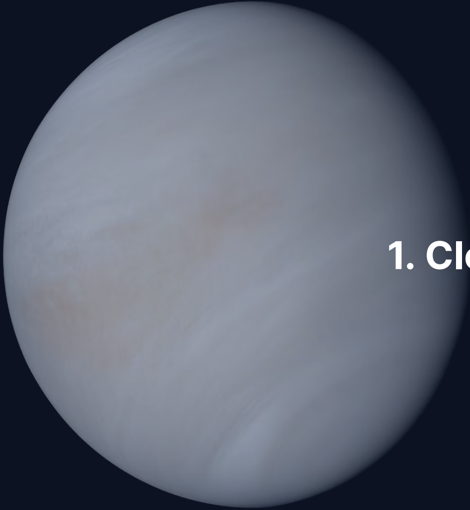
By the end of this lecture, you will be able to:

- Derive the Clausius-Clapeyron equation and apply it to predict cloud formation
- Explain atmospheric circulation using the Coriolis effect and geostrophic balance
- Describe weather phenomena throughout the solar system
- Discuss long-term climate evolution, including the faint young Sun paradox
- Explain the carbonate-silicate cycle as a climate thermostat

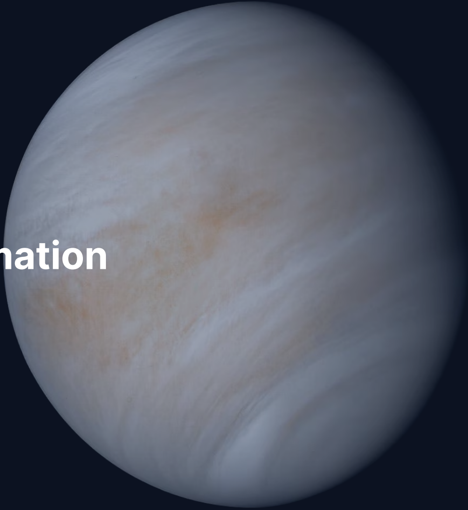
Recap: Lecture 5

- Atmospheric types: primary ($\mathrm{H_2/He}$), secondary (outgassed), tertiary (modified)
- Hydrostatic equilibrium: $\mathrm{d}P = -\rho g\,\mathrm{d}z$
- Scale height: $H = k_\mathrm{B}T/(\mu\, m_u\, g)$
- Radiative equilibrium: $T_\mathrm{eq} = T_\odot(R_\odot/2a)^{1/2}(1-A)^{1/4}$
- Greenhouse effect: surface temperature exceeds equilibrium temperature
- Adiabatic lapse rate: $\Gamma_d = g/c_p$

Today: What happens when vapour condenses? How do winds blow? Why has Earth's climate been stable for 4 Gyr?



1. Cloud formation



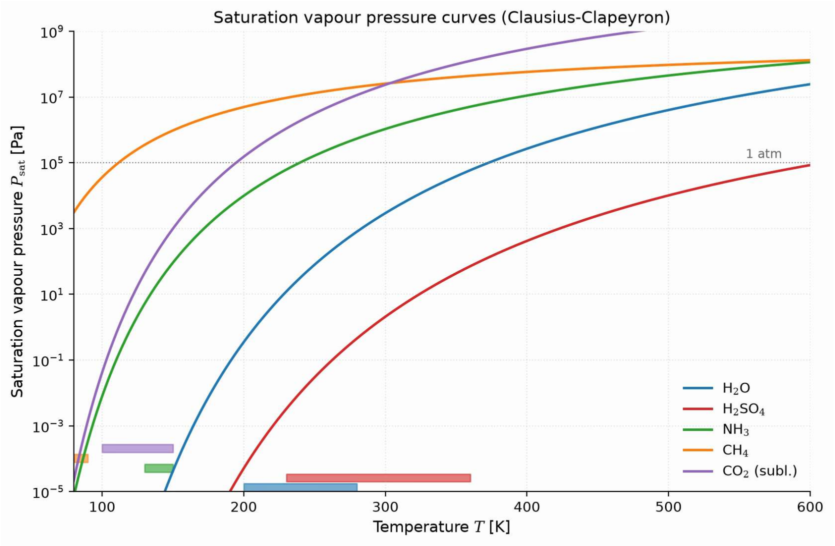
Venus in UV, Mariner 10 (1974). Credit: NASA/JPL-Caltech

Saturation and condensation

- Every vapour has a **saturation vapour pressure** $P_{\mathrm{sat}}(T)$
- When actual vapour pressure exceeds P_{sat}: air is **supersaturated**
- $P_{\mathrm{sat}}(T)$ increases roughly exponentially with temperature
- **Relative humidity:**

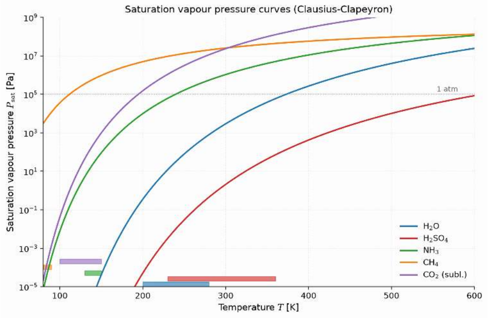
$$\mathrm{RH} = \frac{P_{\mathrm{vapour}}}{P_{\mathrm{sat}}(T)} \times 100\%$$

- $\mathrm{RH} = 100\%$: saturated; $\mathrm{RH} > 100\%$: condensation can occur
- Rising air cools → P_{sat} drops → reaches saturation → clouds



Saturation vapour pressure for the five main condensables. Course figure.

Reading the saturation curves



- Every curve is one Clausius-Clapeyron law: near-exponential rise, steeper for larger L_v/R_v.
- The lanes at the bottom mark where each species condenses: $\mathrm{CH_4}$ at 80–90 K on Titan, $\mathrm{CO_2}$ at 100–150 K on Mars, $\mathrm{H_2O}$ at 200–280 K, $\mathrm{H_2SO_4}$ across Venus's 230–360 K cloud deck.
- For water, a 20 K cooling cuts P_{sat} to a quarter (611 vs. 2360 Pa across 273–293 K): rising air saturates quickly.

A single equation, evaluated per species, predicts every cloud deck in the solar system.

Nucleation: how droplets form

Even in supersaturated air, condensation needs a **seed**.

Homogeneous nucleation: Forming droplets from vapour alone
Requires very high supersaturations ($\mathrm{RH} \gg 100\%$), extremely rare

Heterogeneous nucleation: Condensation onto pre-existing particles
Condensation nuclei: dust, volcanic aerosols, sea salt, soot
Activates at only ~0.04–0.4% supersaturation: the dominant mechanism

- Earth: oceanic and biogenic nuclei
- Mars: wind-lofted mineral dust
- Giant planets: photochemical (UV-driven) hazes from above

The Köhler equation

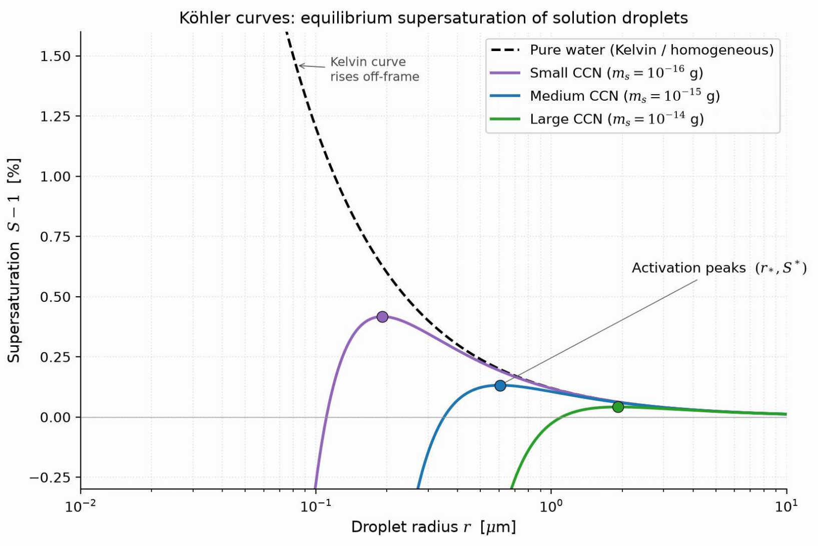
Saturation ratio at which a solution droplet of radius r neither grows nor evaporates:

$$S(r) \approx 1 + \underbrace{\frac{A}{r}}_{\text{Kelvin}} - \underbrace{\frac{B}{r^3}}_{\text{Raoult}}, \qquad A = \frac{2\sigma_w}{\rho_w R_v T}, \qquad B = \frac{3\, i\, M_w\, m_s}{4\pi\rho_w M_s}$$

- **Kelvin (curvature) term** $+A/r$: raises the vapour pressure over a convex surface, because a molecule on a tightly curved droplet has fewer neighbours holding it. Diverges as $r \to 0$, so the smallest embryos demand the most supersaturation.
- **Raoult (solute) term** $-B/r^3$: lowers it, because dissolved solute leaves fewer surface water molecules free to evaporate. The solution concentrates as the droplet shrinks, so this term steepens and dominates at small radius.
- σ_w surface tension, ρ_w liquid density, R_v gas constant of water vapour, i the van 't Hoff factor (ions per formula unit), m_s solute mass, M the molar masses.

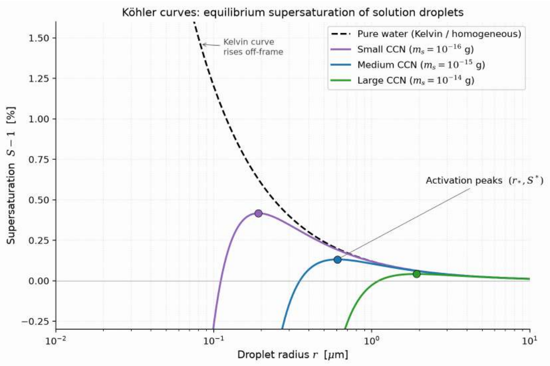
Curvature wins at large r, solute at small r, so $S(r)$ peaks at the **critical radius** r_*: below it a stable haze particle, past it a cloud droplet.

Credit: Catling (2017); Pruppacher & Klett (1997)



Köhler curves: equilibrium supersaturation of solution droplets. Course figure.

Reading the Köhler curves



- The dashed Kelvin curve is the pure-water barrier: ~12% supersaturation at r = 0.01 μm, rising further for smaller embryos.
- Dissolved salt (the Raoult term) pulls the curves down: droplets on condensation nuclei activate at peaks of only 0.04–0.4%.
- Past its activation peak a droplet grows freely: the peaks decide which aerosols become cloud droplets.

Real clouds nucleate heterogeneously: aerosols lower the barrier by two orders of magnitude.

The lifting condensation level (LCL)

- Rising air cools at the dry adiabatic lapse rate $\Gamma_d = g/c_p$
- Vapour pressure stays roughly constant during ascent
- But $P_{\mathrm{sat}}(T)$ decreases as temperature drops
- At the altitude where $P_{\mathrm{vapour}} = P_{\mathrm{sat}}(T)$: condensation begins

The **lifting condensation level** (LCL) marks the **cloud base**. Above it, latent heat release produces the moist adiabatic lapse rate (~5 K km^{-1} in Earth's warm lower troposphere vs. ~9.8 K km^{-1} dry; the observed tropospheric mean is ~6.5).

Latent heat → buoyancy → vigorous convection → thunderstorms, hurricanes, cumulonimbus.

A cumulonimbus tower spreading into a flat **anvil**. Latent heat keeps the rising air warmer than its surroundings and drives the tower up until the stable stratosphere stops it at the tropopause. The anvil top marks where buoyancy ends.

Credit: Eulenjäger, CC BY-SA 3.0, via Wikimedia Commons.

Cloud types of the solar system

What condenses depends on temperature and composition:

Body	Cloud species	Cloud base altitude
Earth	H_2O (liquid & ice)	~1–2 km
Venus	H_2SO_4 (sulfuric acid)	~48 km
Mars	CO_2 ice, H_2O ice	10–100 km
Titan	CH_4, C_2H_6	~8–30 km
Jupiter	NH_3, NH_4SH, H_2O	layered, 0.5–7 bar

The physics is the same everywhere: the **Clausius-Clapeyron equation** governs all of them.

2. Blackboard derivation: the Clausius-Clapeyron equation

Venus in UV. Credit: NASA/JPL-Caltech

Deriving the Clausius-Clapeyron equation

Goal: Derive the exponential dependence of saturation vapour pressure on temperature from thermodynamic phase equilibrium.

Setup: phase equilibrium.

Along the coexistence curve, liquid and vapour are in equilibrium:

$$g_\ell(T, P) = g_v(T, P)$$

where g is the specific Gibbs free energy.

Moving along the coexistence curve by $(\mathrm{d}T, \mathrm{d}P)$:

$$\mathrm{d}g_\ell = \mathrm{d}g_v$$

Derivation: the exact Clausius-Clapeyron relation

From thermodynamics: $\mathrm{d}g = -s\,\mathrm{d}T + v\,\mathrm{d}P$

Setting $\mathrm{d}g_\ell = \mathrm{d}g_v$:

$$(s_v - s_\ell)\,\mathrm{d}T = (v_v - v_\ell)\,\mathrm{d}P$$

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\Delta s}{\Delta v} = \frac{L_v}{T(v_v - v_\ell)}$$

Two approximations:

1. $v_v \gg v_\ell$, so $v_v - v_\ell \approx v_v$
2. Ideal gas: $v_v = R_v T/P$, where $R_v = R^*/M$

$$\frac{\mathrm{d}P_{\mathrm{sat}}}{\mathrm{d}T} = \frac{L_v P_{\mathrm{sat}}}{R_v T^2}$$

Integrating the differential form

Starting from

$$\frac{\mathrm{d}P_{\mathrm{sat}}}{\mathrm{d}T} = \frac{L_v P_{\mathrm{sat}}}{R_v T^2}$$

separate variables:

$$\frac{\mathrm{d}P_{\mathrm{sat}}}{P_{\mathrm{sat}}} = \frac{L_v}{R_v} \frac{\mathrm{d}T}{T^2}$$

Integrate from a reference state $(T_{\mathrm{ref}}, P_{\mathrm{ref}})$ to (T, P_{sat}), assuming $L_v \approx \mathrm{const.}$:

$$\ln\left(\frac{P_{\mathrm{sat}}}{P_{\mathrm{ref}}}\right) = -\frac{L_v}{R_v}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)$$

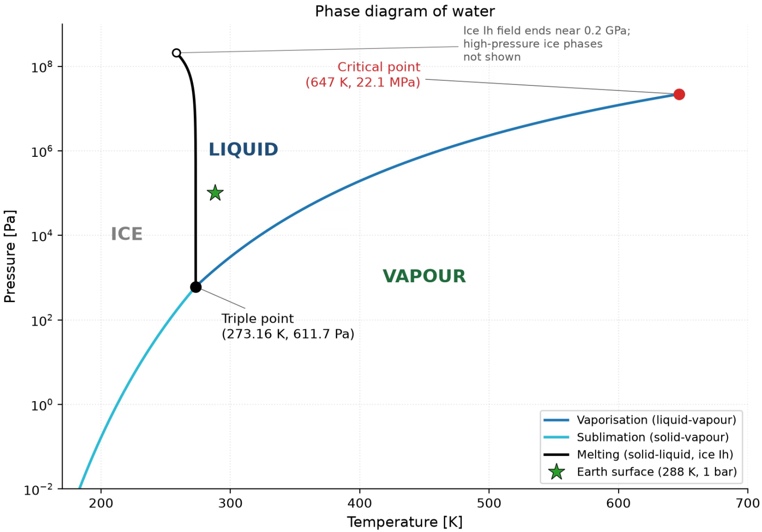
Exponentiate to get the integrated form on the next slide.

The Clausius-Clapeyron equation

$$\boxed{P_{\mathrm{sat}}(T) = P_{\mathrm{ref}} \exp\left[-\frac{L_v}{R_v}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right]}$$

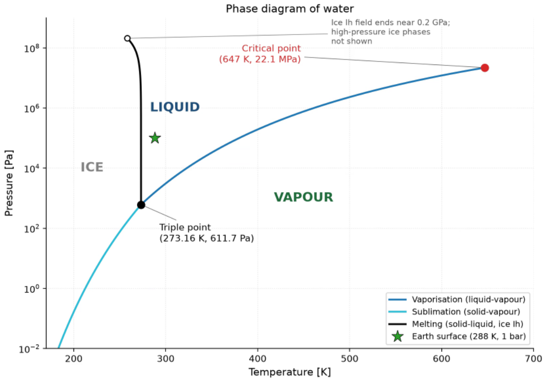
Credit: Derivation follows Catling (2017), *Atmospheric Evolution on Inhabited and Lifeless Worlds*, Ch. 1

- P_{sat} depends **exponentially** on temperature
- The sensitivity is controlled by L_v/R_v (units of K)
- Large L_v/R_v → very sensitive to temperature changes
- This equation applies to **any** vapour ↔ condensed phase transition



Phase diagram of water. Course figure.

The phase diagram behind the equation



- The liquid-vapour curve from the triple point (273.16 K, 611.7 Pa; all phases coexist) to the critical point (647 K, 22.1 MPa) is the integrated Clausius-Clapeyron relation.
- The sublimation branch uses the larger latent heat of sublimation: a steeper curve below the triple point.
- Earth's surface sits inside the liquid field; Mars's ~6 mbar surface lies below the triple-point pressure, so liquid water is not stable there today.

Wherever a *P*–*T* profile crosses the blue curve, the equation predicts a cloud, a rain, or a frost.

Worked example: water on Earth

For water vapour:

L_v	2.50×10^6 J kg^{-1}
M	0.018 kg mol^{-1}
$R_v = R^*/M$	462 J kg^{-1} K^{-1}
$T_{\rm ref}$ (triple point)	273 K
$P_{\rm ref}$ (triple point)	611 Pa
L_v/R_v	$\approx$ 5400 K

At T = 293 K (20°C):

$$P_{\rm sat} = 611 \exp\left[-5400\left(\frac{1}{293} - \frac{1}{273}\right)\right] = 611 \times 3.86 \approx 2360 \text{ Pa}$$

Measured value: 2300 Pa; excellent agreement.

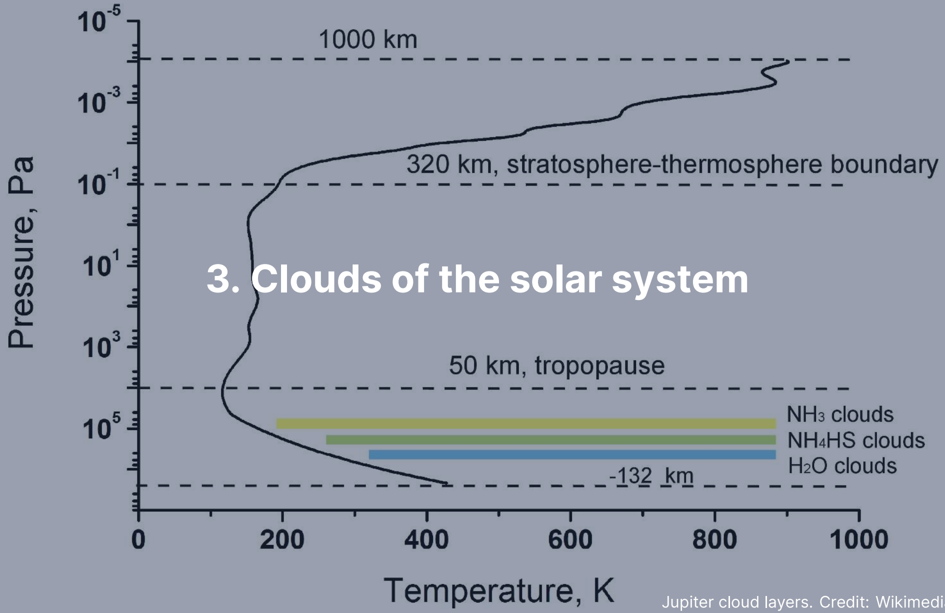
Condensing species in the solar system

Species	L_v (kJ kg^{-1})	R_v (J kg^{-1} K^{-1})	L_v/R_v (K)	T_{cond} (K)	Where
H_2O	2500	462	5400	200–280	Earth, Mars
H_2SO_4	540	85	6400	230–360	Venus
NH_3	1370	488	2800	130–150	Jupiter, Saturn
CH_4	510	519	980	80–90	Titan, Uranus, Neptune
CO_2	571	189	3020	100–150	Mars

Source: Catling (2017), de Pater & Lissauer (2010); T_{cond} is the approximate condensation range at each body's pressure levels

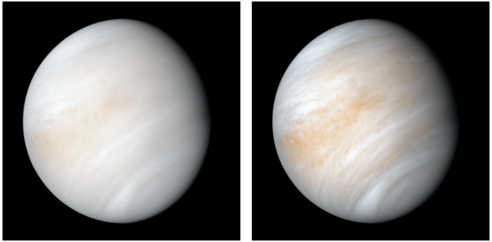
- High L_v/R_v for H_2SO_4: steep saturation curve, sharply defined cloud base on Venus
- Low L_v/R_v for CH_4: flat curve, Titan stays near saturation over a wide altitude range

3. Clouds of the solar system



Jupiter cloud layers. Credit: Wikimedia, CC BY-SA 3.0

Venus: sulfuric acid clouds

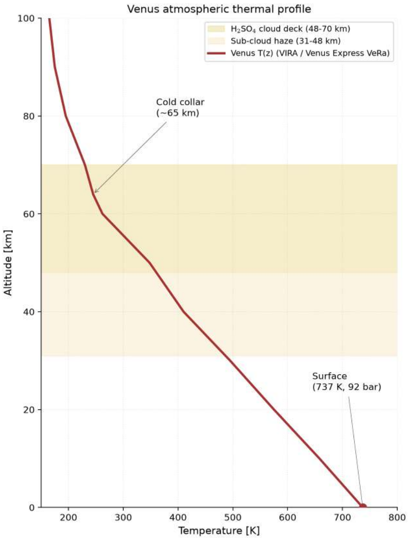


Credit: NASA/JPL-Caltech (1974)

- H_2SO_4 clouds from 48 to 70 km altitude
- Completely obscure the surface at visible wavelengths
- Photochemistry:

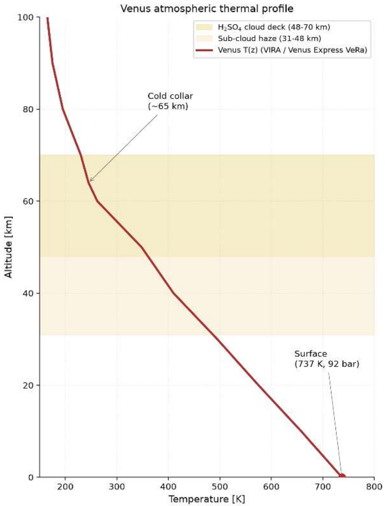
$$SO_2 + O \rightarrow SO_3, \quad SO_3 + H_2O \rightarrow H_2SO_4$$

- SO_2 supplied by volcanic outgassing; O from CO_2 photolysis
- Unidentified **UV absorber** creates banded patterns
- Sub-cloud haze extends down to ~31 km



Venus thermal structure, VIRA / Venus Express VeRa. Course figure.

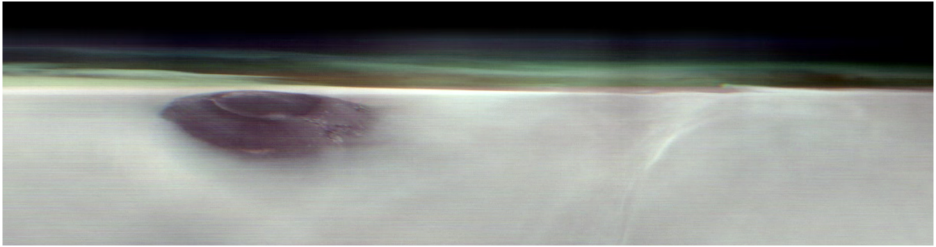
Where the Venus clouds live



- The deck occupies 48–70 km, where *T* runs from ~360 K at the base to ~230 K at the top: the stability window of concentrated H_2SO_4 droplets.
- Below it: a thin haze down to ~31 km, then clear, oven-hot CO_2 to the 737 K, 92 bar surface.
- The cold-collar inversion near 65 km shapes the cloud top at high latitudes.

The clouds sit where Clausius-Clapeyron allows them: a 22 km deck bounded by evaporation below and photochemical H_2SO_4 supply above.

Mars: dust and ice clouds



Water-ice clouds over Tharsis at dawn, seen along the limb. The dark peak breaking through the canopy is Arsia Mons. Credit: NASA/JPL-Caltech/ASU (Mars Odyssey THEMIS, 2 May 2025)

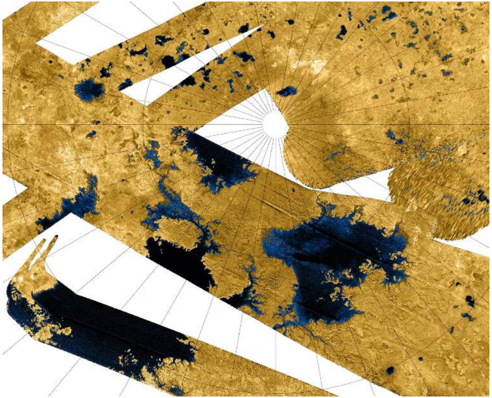
Mars's thin atmosphere (~6 mbar) supports two cloud types:

- CO_2 **ice clouds:** 50–100 km; frost point ~148 K at the 6 mbar surface, ~100 K at mesospheric pressure
- H_2O **ice clouds:** 10–30 km, over Tharsis, aphelion cloud belt

Mineral dust plays a central role:

- Condensation nuclei for ice clouds
- Absorbs solar radiation, heats atmosphere
- More dust → more heating → stronger winds → more dust

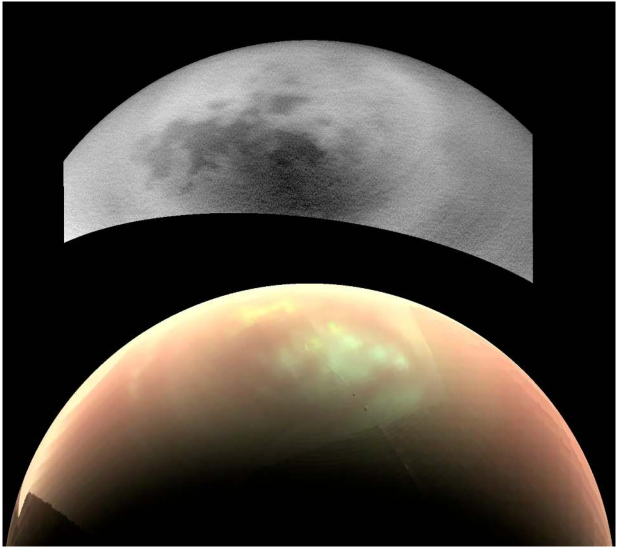
Titan's methane cycle: an Earthlike analogue



Credit: NASA/JPL-Caltech/ASI/USGS (Cassini)

Titan has the only other active hydrological cycle in the solar system, with methane replacing water.

- **Surface:** liquid methane seas near the north pole
- **Kraken Mare:** ~400,000 km^2 (Caspian-sized), locally >100 m deep (Hayes 2016; Poggiali et al. 2020)
- Rivers carve channels; methane rain fills lakebeds
- Evaporation at equator, rain at poles
- 30-yr Titan year drives slow seasonal cycling
- Cassini radar mapped lakes; Dragonfly will sample in 2034+

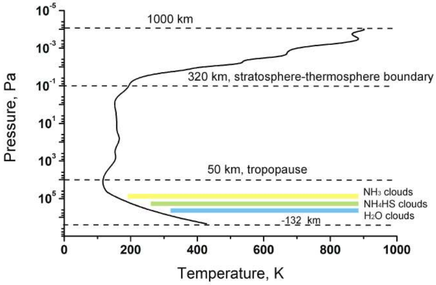


Methane clouds at Titan’s mid-southern latitudes, Cassini ISS. Credit: NASA/JPL-Caltech/SSI.

Titan: methane rain

- Only known active **hydrological cycle** beyond Earth, with $\mathrm{CH_4}$ replacing $\mathrm{H_2O}$
- Surface: $T \approx 94$ K, $P \approx 1.5$ bar, near the triple point of methane
- Liquid methane: lakes, seas, rivers carved into the icy surface
- $\mathrm{CH_4}$ clouds at 8–30 km; $\mathrm{C_2H_6}$ (ethane) clouds also present
- Rainfall infrequent but intense
- Seasonal cloud migration observed by Cassini–Huygens

Giant planets: layered cloud structure



Credit: Wikimedia, CC BY-SA 3.0

Vertically layered, predicted by Clausius-Clapeyron:

1. $\mathrm{NH_3}$ **ice** (top): *T* ~ 130–150 K, *P* ~ 0.5–1 bar
   The visible white and coloured bands
2. $\mathrm{NH_4SH}$: *T* ~ 200–240 K, *P* ~ 2–3 bar
   $\mathrm{NH_3} + \mathrm{H_2S} \rightarrow \mathrm{NH_4SH}$
3. $\mathrm{H_2O}$ **ice/liquid** (deep): *T* ~ 270–300 K, *P* ~ 5–7 bar
   Drives weather via latent heat release

Ice giants: $\mathrm{CH_4}$ on top, $\mathrm{H_2S}$ below, $\mathrm{NH_4SH}$ and $\mathrm{H_2O}$ deeper.

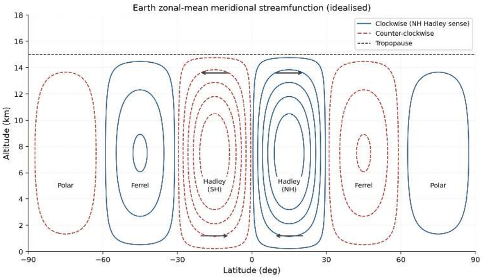
Break

Earthrise, Apollo 8 (1968). Credit: NASA

4. Atmospheric dynamics

Jet-stream cirrus from orbit. Credit: NASA/JSC

The Hadley cell

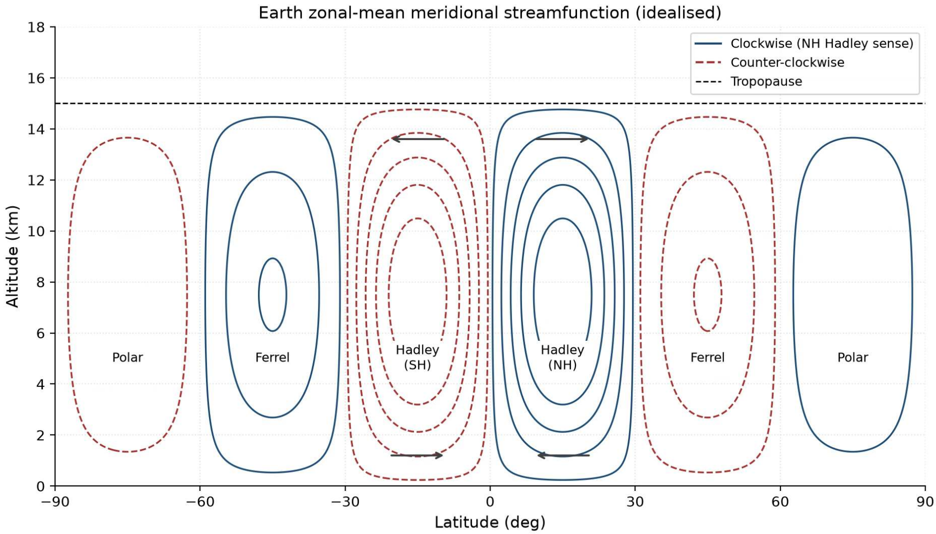


Credit: Course figure

1. Intense solar heating at equator → air rises
2. At altitude, air flows poleward
3. Air cools radiatively, sinks at ~30° latitude
4. Surface return flow → **trade winds**

This loop is the **Hadley cell**:

- Dominant circulation in the tropics
- Primary heat transport: equator → poles
- Sinking zones → desert belts on Earth



Zonal-mean (averaged along latitude circles) overturning circulation of Earth's troposphere. Course figure.

The Coriolis effect

On a rotating planet, moving air is **deflected**:

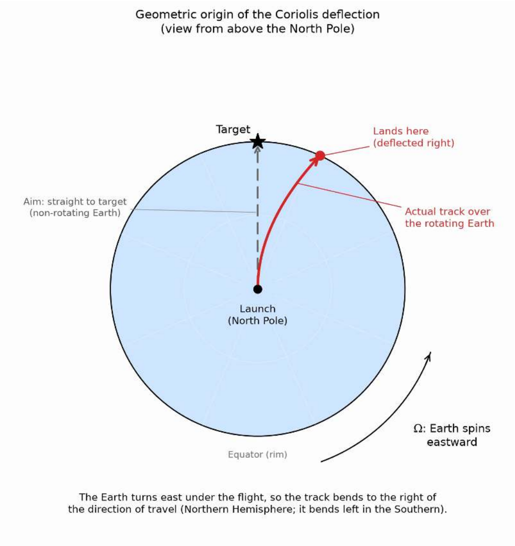
- **Right** in the Northern Hemisphere
- **Left** in the Southern Hemisphere

The **Coriolis parameter**:

$$f = 2\Omega \sin \phi$$

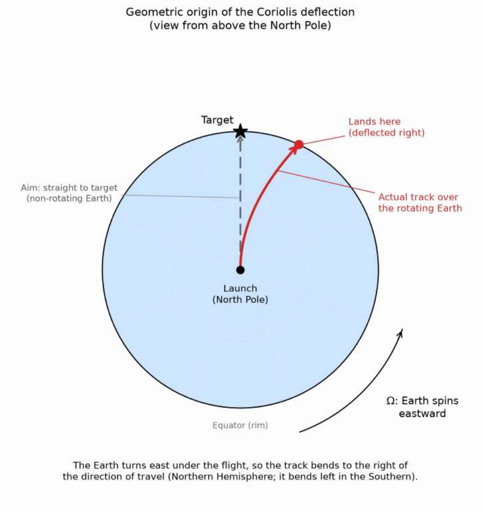
- Earth: $\Omega = 7.29 \times 10^{-5}$ rad s^{-1} (sidereal)
- At mid-latitudes ($\phi = 45°$): $f \approx 1.03 \times 10^{-4}$ s^{-1}
- At equator ($\phi = 0°$): $f = 0$, so no Coriolis deflection

The Coriolis effect explains why winds do not blow directly from high to low pressure, and why weather systems rotate.



Coriolis deflection of a polar launch, viewed from above the North Pole. Course figure.

Where the deflection comes from



- A projectile launched from the pole has no eastward velocity: on a non-rotating planet it would fly straight to its target (dashed line).
- The planet turns east under the flight, so the track over the ground curves right of the direction of travel and lands right of the target.
- Launching from the pole isolates the effect. Strength is set by $f = 2\Omega \sin \phi$: zero at the equator, largest at the poles, mirrored left in the Southern Hemisphere.

No force pushes the parcel sideways. The deflection is the rotating planet turning beneath a straight path.

The Rossby number

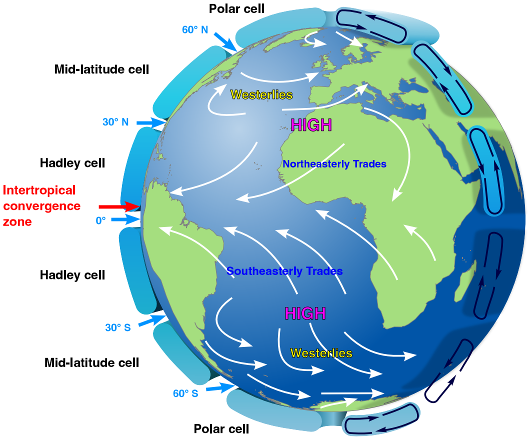
Does rotation matter for a given flow?

$$\mathrm{Ro} = \frac{U}{fL}$$

$\mathrm{Ro} \ll 1$: Rotation dominates: large-scale circulation, jet streams
Earth: $U \sim 10\ \mathrm{m\,s^{-1}}$, $L \sim 1000$ km $\rightarrow \mathrm{Ro} \sim 0.1$

$\mathrm{Ro} \gg 1$: Rotation unimportant: tornadoes, dust devils

Determines the number of circulation cells on a planet.



The three-cell circulation of a moderately rotating planet: the Hadley cell (equator to ~30°), the mid-latitude or **Ferrel** cell (~30–60°), and the polar cell. The northeasterly and southeasterly trades and the westerlies are their surface expression. Credit: Wikimedia Commons, public domain.

Circulation cells and rotation rate

Slowly rotating (Venus, Titan):

Single Hadley cell from equator to pole in each hemisphere.
Venus: 243-day period → $\mathrm{Ro} \gg 1$ for large-scale flows.

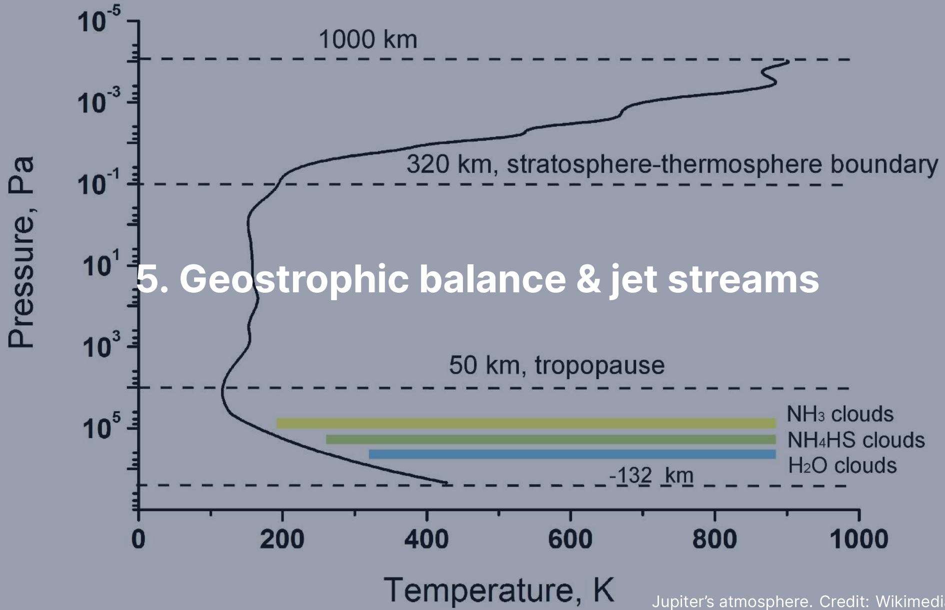
Moderately rotating (Earth):

Hadley cell to ~30°, then Ferrel cell (mid-latitudes), polar cell.
Three cells per hemisphere.

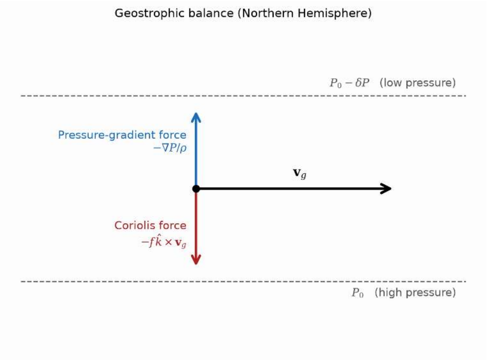
Rapidly rotating (Jupiter, Saturn):

Many alternating cells → **banded structure**.
Jupiter: many alternating jets per hemisphere, out to ~ ±60°.
Light zones (rising air, high clouds) and dark belts (sinking air).

5. Geostrophic balance & jet streams



Geostrophic balance



At large scales ($\mathrm{Ro} \ll 1$), steady state = Coriolis force balances pressure gradient:

$$f\,\hat{k} \times \mathbf{v}_g = -\frac{1}{\rho}\nabla P$$

In scalar form:

$$u_g = -\frac{1}{f\rho}\frac{\partial P}{\partial y}, \qquad v_g = \frac{1}{f\rho}\frac{\partial P}{\partial x}$$

The geostrophic wind blows **parallel to isobars** (constant-pressure lines), not high to low. In the Northern Hemisphere: low pressure to the left of the wind.

This is why cyclones and anticyclones **rotate** around pressure centres.

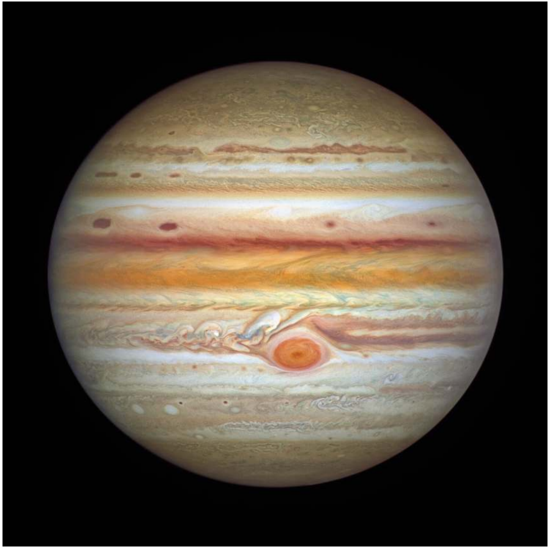
Jet streams and the thermal wind

Jet streams: Narrow bands of fast winds (~30–70 m s^{-1} on Earth).

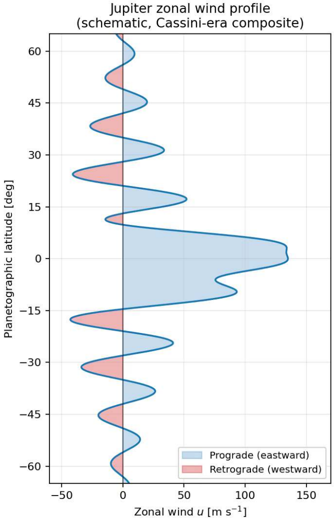
The **thermal wind relation** connects vertical wind shear to horizontal temperature gradients:

$$\frac{\partial \mathbf{v}_g}{\partial z} \propto \hat{k} \times \nabla T$$

- Where ∇T is steepest (tropics vs. poles boundary): wind speed increases with altitude
- On Earth: subtropical jet (~30°) and polar jet (~60°)
- Jet streams **steer weather systems** across the planet

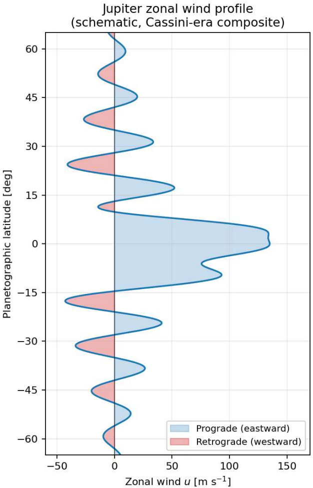


Full-disk Jupiter, HST/OPAL, 4 September 2021. Credit: NASA, ESA, A. Simon (GSFC), M. H. Wong (UC Berkeley) and the OPAL team; ESA/Hubble, CC BY 4.0.



Jupiter's cloud-top zonal (east-west) winds, schematic after García Melendo & Sánchez-Lavega (2001). Course figure.

Reading Jupiter's jet ladder



- Alternating prograde (eastward, blue) and retrograde (westward, red) jets stack from the fast eastward equatorial jet out to ~±60° latitude.
- The jets sit at the boundaries between bright zones and dark belts; peak speeds reach ~180 m s^{-1}.
- Juno gravity data: best-fit e-folding (1/e-decay) depth ~1800 km (range 1000–3000 km) into the molecular envelope.

How deep do the bands go? Juno's gravity data put the jets thousands of kilometres into the envelope.

Giant planet banding

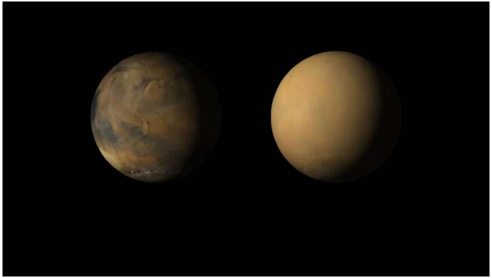
- **Zones** (light): rising air, high clouds (ammonia ice tops)
- **Belts** (dark): sinking air, deeper cloud layers exposed
- Strong **zonal jets** between adjacent bands:
  - Jupiter: peaks of ~180 $m\,s^{-1}$
  - Saturn: ~400 $m\,s^{-1}$
- Jets remarkably stable over decades of observation
- Juno gravity: best-fit decay depth ~1800 km into the molecular envelope

The same geostrophic and thermal wind physics that produces Earth's jet streams creates the banded structure of the giant planets, just with many more cells due to rapid rotation.

6. Weather & storms in the solar system

Jupiter’s Great Red Spot. Credit: NASA/JPL-Caltech/SwRI/MSSS

Mars: dust storms



Credit: NASA/JPL-Caltech/MSSS

- Local storms: dust lofted to 10–30 km
- Occasionally grow into **global dust storms**
- 2018 global storm ended *Opportunity*'s 15-year mission
- **Positive feedback loop:** dust absorbs sunlight → heats air → stronger winds → more dust

Seasonal $\mathrm{CO_2}$ cycle:

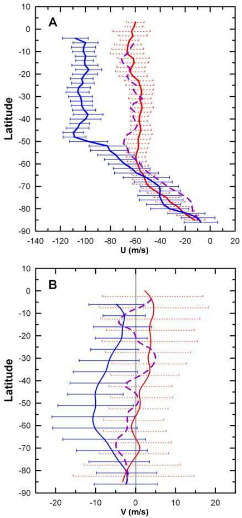
- ~25–30% of atmosphere freezes onto polar caps each winter
- Surface pressure swings: 5–7 mbar

Venus: super-rotation

- Venus's surface rotates once every 243 Earth days (retrograde)
- Cloud-top winds: ~100 m s^{-1}, circling the planet in ~4 days
- Atmosphere rotates **60× faster than the planet**
- This is **atmospheric super-rotation**

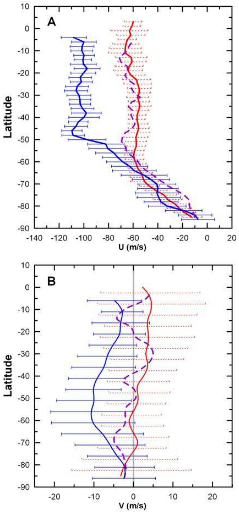
A major puzzle

Super-rotation requires angular momentum transport **upward and equatorward**, against friction. Leading explanation: thermal tides + planetary-scale waves. The detailed mechanism remains an active area of research.



Venus cloud-level winds from VIRTIS on Venus Express: zonal (A) and meridional (B) speed against latitude. Each colour is one wavelength, so one altitude: **blue** 380 nm (~66 km), **violet dashed** 980 nm (~61 km), **red** 1.74 μm (~47 km). Sánchez-Lavega et al. (2008).

Measuring super-rotation



- Each colour is one sounding wavelength, so one altitude: **blue** 380 nm, ~66 km; **violet dashed** 980 nm, ~61 km; **red** 1.74 μm, ~47 km.
- Zonal winds: ~105 m s^{-1} at the cloud top, 60–70 m s^{-1} at the cloud base: the whole deck super-rotates.
- Meridional winds are ten times weaker, poleward at the top: a slow overturning beneath a fast zonal spin.

The super-rotation extends through the whole cloud deck, 60 times faster than the planet beneath.

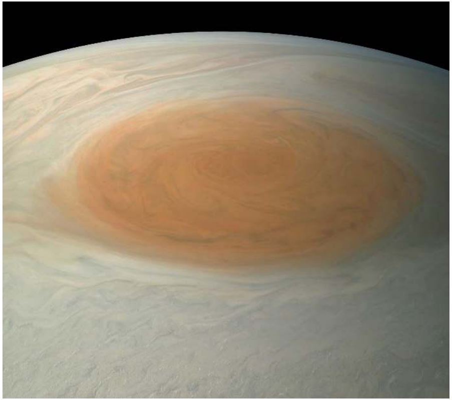
Venus's super-rotation: thermal tides

How does Venus's atmosphere rotate 60 times faster than the surface?

- Surface rotation: 243 days, retrograde
- Cloud-top winds: 4 days, ~100 m/s
- Sustaining this requires angular momentum transport **upward and equatorward**
- **Thermal tides:** solar heating of the cloud layer creates day-night pressure gradients, which launch waves that carry angular momentum upward
- **Planetary-scale waves** redistribute angular momentum toward the equator
- Detailed mechanism still debated; Akatsuki measurements ongoing

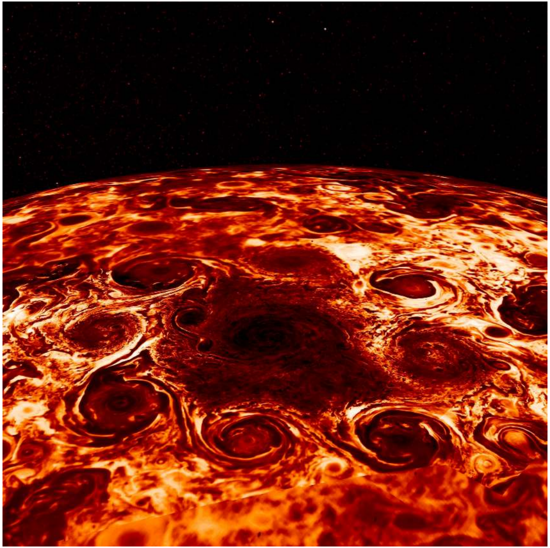
Source: Sánchez-Lavega et al. (2017)

Jupiter: the Great Red Spot



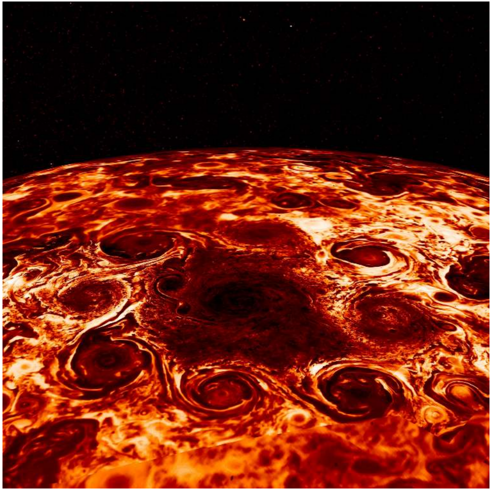
Credit: NASA/JPL-Caltech/SwRI/MSSS (Juno)

- **Largest storm in the solar system**
- Anticyclonic vortex, larger than Earth
- Winds: ~120 $\mathrm{m\,s^{-1}}$ at periphery
- First telescopic sightings 1664–65 (possibly a different spot); uninterrupted records since ~1830
- Confined between two opposing zonal jets
- Sustained by absorbing smaller vortices + deep $\mathrm{H_2O}$ latent heat
- Slowly **shrinking** over the past century



Jupiter’s north-polar cyclone cluster from JIRAM, Juno’s infrared imager. Credit: NASA/JPL-Caltech/SwRI/ASI/INAF.

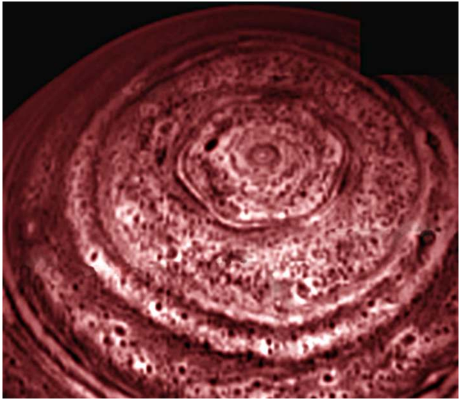
Polar order instead of polar chaos



- Eight cyclones ring a central one at Jupiter's north pole; five ring the south pole.
- The clusters have stayed in place over years of Juno perijoves (closest-approach flybys): no merging, no drifting apart.
- A sixth southern cyclone appeared in late 2019 and dissipated within about two months (Mura et al. 2021).

Stable polygons of cyclones: a regime with no Earth analogue, probing how deep rotating convection organises itself.

Saturn: the hexagonal jet stream



Credit: NASA/JPL-Caltech/SSI (Cassini, 2013)

- Persistent hexagonal pattern at ~78°N
- Diameter: ~30,000 km (> Earth's diameter)
- First seen by Voyager (1980–81), later imaged by Cassini
- Explained as a stable **Rossby wave** (Coriolis-restored) with six-fold symmetry
- Resonance between jet speed, width, and Coriolis gradient

Saturn also has periodic **Great White Storms** (~every 30 yr; last: 2010).

Neptune’s Great Dark Spot, Voyager 2 (1989). Credit: NASA/JPL-Caltech.

Neptune: extreme weather on a cold world

- Receives only ~1/900th of Earth's solar flux
- Yet has the **fastest winds** in the solar system: ~580 $m\,s^{-1}$
- Voyager 2 (1989): Great Dark Spot (since vanished), new storms formed
- Storms more transient than Jupiter's

Neptune radiates ~2.6× more energy than it receives from the Sun. Internal heat (gravitational contraction) drives convection and storms even with negligible solar heating.

A powerful reminder that solar flux alone does not determine weather.

7. Climate evolution & the faint young Sun

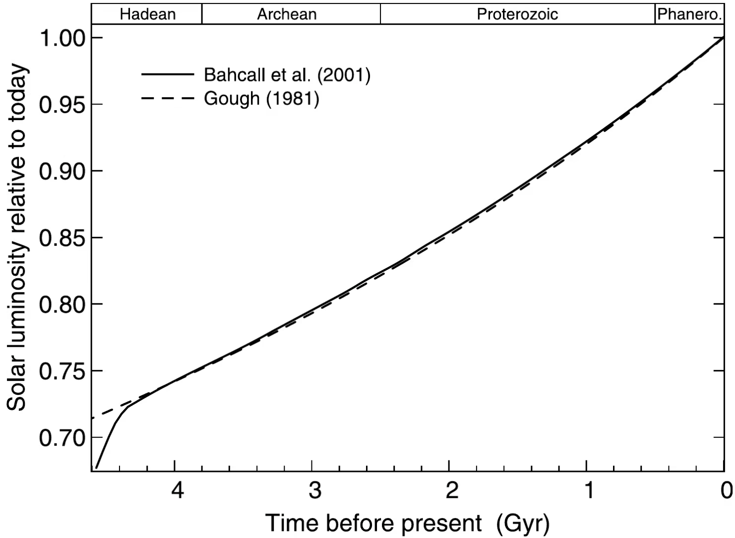
Mars global dust storm (2018). Credit: NASA/JPL-Caltech/MSSS

Solar luminosity evolution

The Sun brightens as H → He increases the core mean molecular weight:

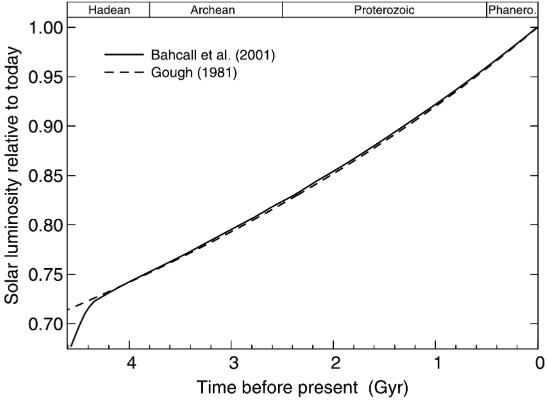
$$\frac{L(t)}{L_\odot} \approx \left[1 + \frac{2}{5}\left(1 - \frac{t}{t_\odot}\right)\right]^{-1}$$

- $t_\odot \approx 4.57$ Gyr (present age of the Sun)
- At formation ($t = 0$): $L/L_\odot \approx 0.71$
- The early Sun was ~**30% fainter** than today
- Even 4 Ga: still ~25% fainter



Solar luminosity across the geologic eons. Reproduced from Feulner (2012).

Reading the faint-young-Sun curve



- The standard solar model and Gough's analytic formula agree to ~0.1% over the past 4 Gyr.
- At the start of the Archean the Sun was ~25% fainter: with today's greenhouse, Earth would have been frozen.
- The eon labels along the top tie the luminosity curve to the geological record: liquid water is inferred throughout.

The Sun brightened by a third over Earth's lifetime, yet the sedimentary record never freezes over. Something compensated.

The faint young Sun paradox

A 30% fainter Sun → $T_{\mathrm{eff}} \approx 255 \times (0.71)^{1/4} \approx 234$ K

With a modern greenhouse: surface well below freezing → **frozen ocean**.

But the geological record says otherwise:

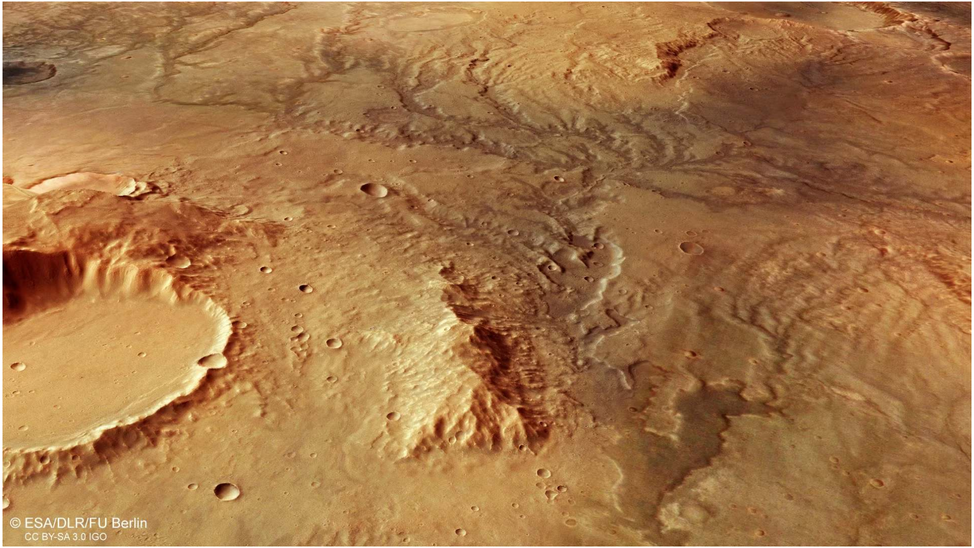
- **Zircon crystals** (4.4 Ga): $\delta^{18}\mathrm{O}$ ratios require liquid water
- **Pillow basalts and sedimentary rocks** (3.8 Ga): formed underwater
- **Stromatolites** (3.5 Ga): microbial mats requiring warm liquid water

Faint Sun should have frozen Earth, but geological evidence demands liquid water for >4 Gyr. This is the **faint young Sun paradox** (Sagan & Mullen, 1972).

Possible solutions

- **Enhanced CO_2 greenhouse:** 10–1000× more CO_2 than today
  Plausible: less exposed continental area, so the weathering thermostat settles at higher CO_2
- **Methane greenhouse:** Biogenic CH_4 from early methanogens
  Long-lived in an anoxic (O_2-free) atmosphere
- **N_2 pressure broadening:** Thicker N_2 atmosphere (2–3× present)
  Enhances CO_2 and H_2O absorption bands
- **Lower albedo:** Less land area + fewer clouds → absorb more sunlight

Most likely: **combination** of elevated CO_2 and CH_4, regulated by the carbonate-silicate cycle.



Noachian valley networks, Mars Express HRSC. Credit: ESA/DLR/FU Berlin, CC BY-SA 3.0 IGO.

The Mars climate puzzle

Mars at 1.52 AU → less than half of Earth's solar flux. With a faint young Sun: even worse.

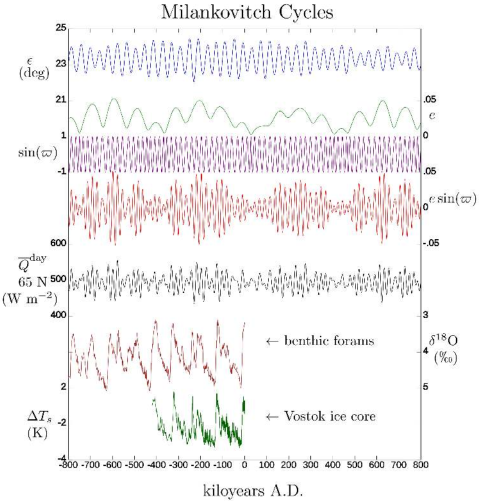
Yet the geological evidence for warm, wet conditions in the Noachian (>3.7 Ga) is strong:

- Extensive valley networks carved by flowing water
- Clay minerals from aqueous weathering
- Sedimentary deposits in ancient lake basins (e.g., Jezero crater)

Dense CO_2 alone is insufficient: it condenses into ice clouds that *increase* albedo.

Proposed solutions: Reducing gases (H_2, CH_4) from volcanism and water-rock reactions. Early Mars climate remains a **major open problem**.

Milankovitch cycles: orbital forcing



Credit: Wikimedia, CC BY-SA

- **Obliquity** (axial tilt): 22.1°–24.5° on ~41 kyr
- **Eccentricity:** e = 0–0.06 on 100 and 400 kyr
- **Climatic precession** (spin-axis wobble plus the turning orbital ellipse): 19–23 kyr
- Change seasonal/latitudinal insolation
- Pace glacial-interglacial cycles
- Match marine δ^{18}O records

Source: Hays et al. (1976); Laskar et al. (2004)

Climate feedbacks

Climate stability depends on feedback mechanisms:

Ice-albedo feedback (positive):

Cooling → ice expands → higher albedo → less absorption → more cooling.
Can trigger **snowball Earth** (Neoproterozoic; see later).

Water vapour feedback (positive):

Warming → more evaporation → $\mathrm{H_2O}$ is a greenhouse gas → more warming.
Largest positive feedback: CO_2 doubling warms ~3 K with feedbacks, ~1.2 K without (IPCC AR6). If runaway: **Venus scenario**.

Cloud feedback (complex):

Low clouds reflect sunlight (cooling); high cirrus traps IR (warming).
Net effect: largest uncertainty in climate models.

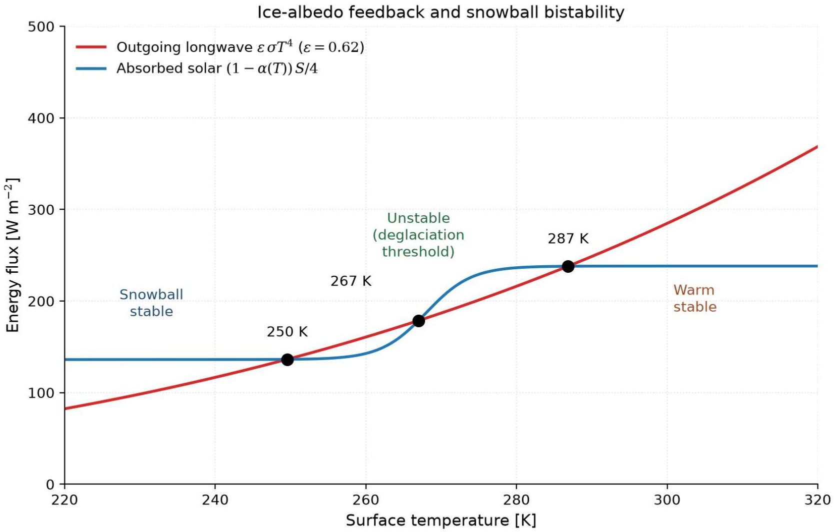
Ice-albedo feedback: the ultimate amplifier

A positive feedback loop that can destabilise climate:

T drops → ice cover grows → albedo rises → less solar absorption → T drops further

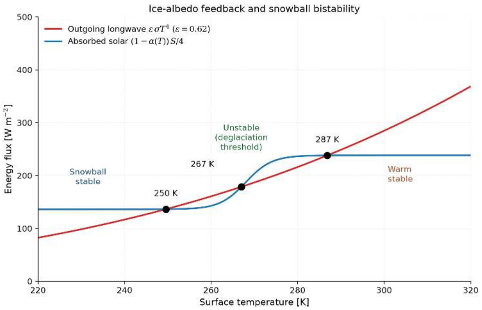
- Ice and snow reflect ~60–90% of incident sunlight vs. ~10–30% for ocean and land
- For Earth: the feedback is strong enough to create a **tipping point**
- If the ice line advances to ~30° latitude (about half the surface area), the feedback runs away
- Runaway cooling then drives the planet into a **snowball state**
- On modern Earth the same feedback amplifies polar warming (Arctic amplification), far from the runaway threshold

Source: Pierrehumbert (2010); North et al. (1981)



Energy-balance bistability of the ice-albedo feedback. Course figure.

Reading the bistability diagram



- Red: outgoing longwave $\varepsilon\sigma T^4$; blue: absorbed sunlight, albedo 0.3 (ice-free) to 0.6 (ice-covered).
- Three crossings: snowball 250 K, threshold 267 K, warm state 287 K.
- The steeper curve decides stability: red steeper at 250 and 287 K pulls the planet back (stable); blue steeper at 267 K amplifies any nudge (unstable).
- Past the middle crossing the feedback carries the climate the rest of the way.

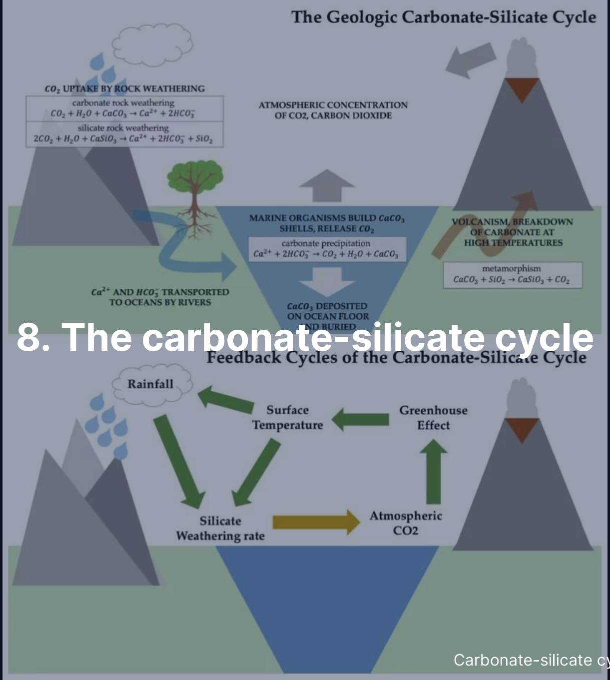
Same Sun, two stable Earths: which one you get depends on history.

Snowball Earth

- **Two snowball states** in the Neoproterozoic (~717 and 635 Ma)
- Evidence: glacial deposits laid down at tropical latitudes
- Triggered by ice-albedo runaway when the ice line reaches ~30° latitude
- Surface temperatures drop below −50°C
- Ice shuts down silicate weathering, so volcanic CO_2 builds up unchecked over ~10–100 Myr
- Greenhouse overwhelms albedo; ice melts rapidly
- Correlated with the rise of complex life soon after

Source: Hoffman et al. (1998); Hoffman et al. (2017)

8. The carbonate-silicate cycle



Carbonate-silicate cycle. Credit: Wikimedia, CC BY-SA 4.0

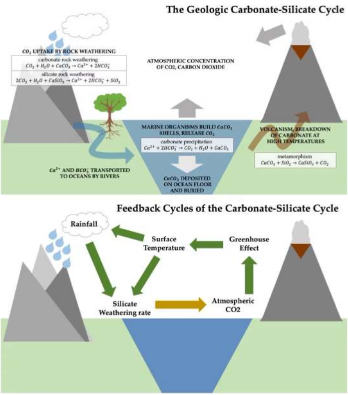
The Urey reaction

Silicate weathering by atmospheric $\mathrm{CO_2}$ dissolved in rainwater:

$$\mathrm{CaSiO_3 + CO_2 + H_2O \longrightarrow CaCO_3 + SiO_2 + H_2O}$$

- $\mathrm{CO_2}$ dissolves in rain → weak acid
- Acid reacts with silicate minerals at the surface
- Products: limestone ($\mathrm{CaCO_3}$) and silica ($\mathrm{SiO_2}$)
- Transported by rivers to the ocean → carbonate precipitates (shells and skeletons, or abiotically) → sediment
- Net effect: **draws** $\mathrm{CO_2}$ **out of the atmosphere** (carbon sink)

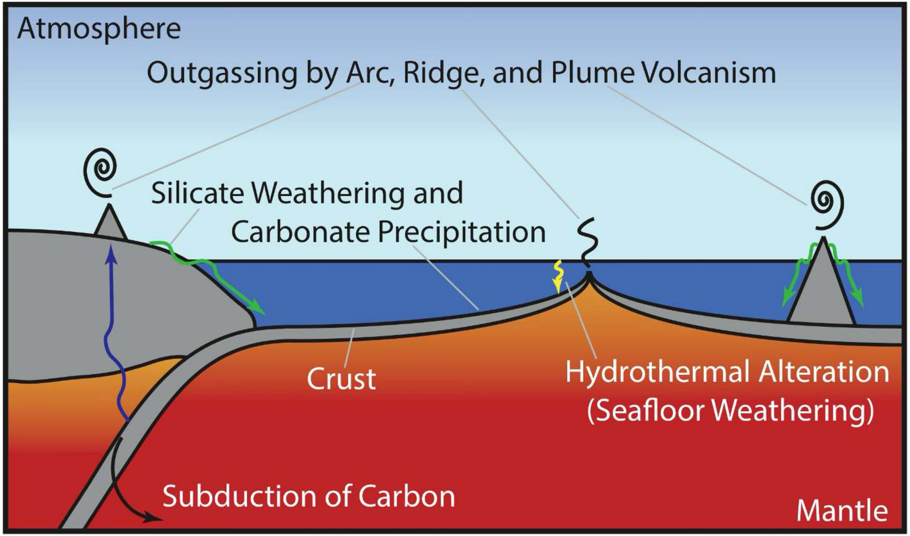
Volcanic outgassing: the carbon source



Credit: Wikimedia, CC BY-SA 4.0

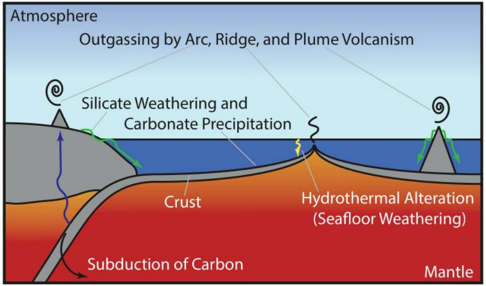
- **Plate tectonics** closes the cycle
- Carbonate ocean floor is subducted into the mantle
- High *T* and *P* decompose carbonates → release CO_2
- CO_2 returned to the atmosphere via **volcanic outgassing**
- This is the long-term carbon **source**

The cycle balances sink (weathering) against source (volcanism).



The carbonate-silicate cycle on the plate-tectonic Earth. Reproduced from Foley (2024).

Sources and sinks of the thermostat



- Sources: arc, ridge, and plume volcanism return $\mathrm{CO_2}$ from the mantle and from subducted carbonates.
- Sinks: silicate weathering and carbonate burial, with seafloor weathering as a secondary sink.
- Warmer climate → faster weathering → $\mathrm{CO_2}$ drawdown → cooling; response time ~0.5 Myr.

A closed loop through atmosphere, ocean, crust, and mantle: break any link (Venus, Mars) and the thermostat fails.

The negative feedback thermostat

Weathering rate depends strongly on temperature (Arrhenius-type):

1. **Planet warms:**
   More rainfall + faster reactions → weathering increases
   → more $\mathrm{CO_2}$ removed → greenhouse weakens → **cools down**
2. **Planet cools:**
   Less rainfall + slower reactions → weathering decreases
   → volcanic $\mathrm{CO_2}$ accumulates → greenhouse strengthens → **warms up**

This thermostat operates on ~0.5 Myr timescales. It is the primary reason Earth has maintained habitable temperatures for >4 Gyr despite a 30% increase in solar luminosity.

Quantifying the weathering feedback

A simple model for the carbonate-silicate thermostat:

$$W(T) = W_0 \exp\left(\frac{T - T_0}{T_e}\right)$$

where W is the weathering rate, $T_e \approx 10$ K is the e-folding temperature.

- Doubling weathering requires ~7 K warming
- Halving it requires ~7 K cooling
- In steady state, $W = V$ (volcanism) sets atmospheric pCO_2
- Perturbation analysis: climate damping timescale ~0.5 Myr
- Works only if liquid water and active volcanism coexist

Source: Walker et al. (1981); Pierrehumbert (2010)

Venus & Mars: failed thermostats

The cycle requires **liquid water** (to weather rocks) and **active volcanism** (to recycle carbon):

Venus:

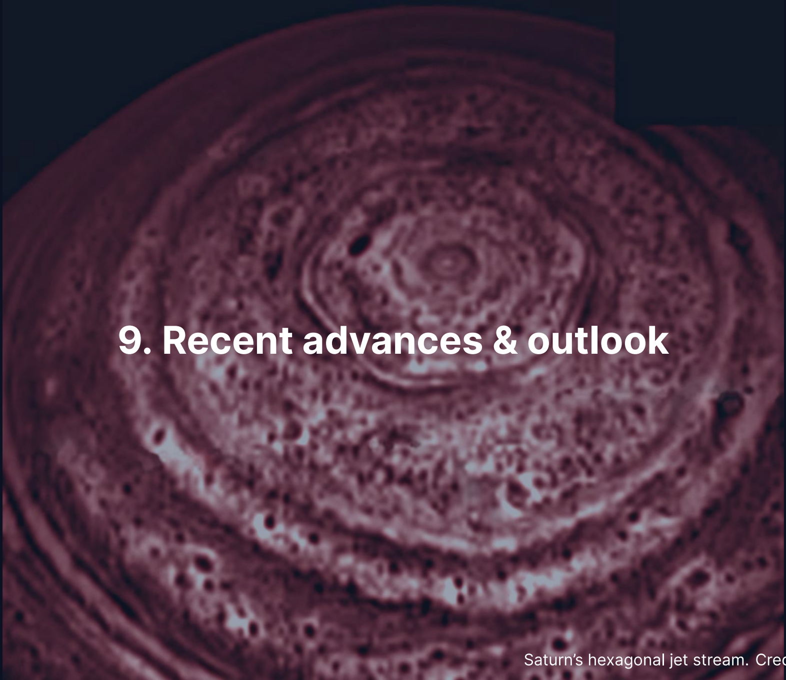
- Lost surface water (runaway greenhouse + H escape)
- No water → no silicate weathering
- Volcanic $\mathrm{CO_2}$ accumulated unchecked
- Result: 92 bar $\mathrm{CO_2}$ atmosphere

Mars:

- Interior cooled, volcanism ceased (~3 Ga)
- No volcanic $\mathrm{CO_2}$ source
- Remaining $\mathrm{CO_2}$ drawn down by weathering + lost to space
- Result: thin 6 mbar atmosphere

Habitability requires not just the right temperature, but the right **geological activity** to sustain it over billions of years.

9. Recent advances & outlook



Saturn’s hexagonal jet stream. Credit: NASA/JPL-Caltech/SSI

The runaway greenhouse

The water-vapour feedback has a limit, and past it a planet loses its oceans.

- Outgoing longwave radiation cannot rise without bound as the surface warms: there is a maximum, and above it the feedback no longer closes
- **Lecture 9** derives that maximum and applies it to Venus

The Venus phosphine debate

In 2020, a team reported tentative detection of phosphine (PH_3) in Venus's clouds.

- Greaves et al. (2021): ~20 ppb PH_3 at 53–60 km altitude
- No known abiotic source in Venus's atmosphere could explain this abundance
- Sparked biological speculation: could PH_3 be produced by hypothetical Venusian microbes?
- Subsequent reanalyses: signal much weaker or absent (Villanueva et al. 2021); mesospheric SO_2 reproduces the feature (Lincowski et al. 2021)
- Current status: unconfirmed, controversial
- Motivated new Venus missions: DAVINCI (NASA), EnVision (ESA), VERITAS (NASA)

Source: Greaves et al. (2021), *Nat. Astron.*; Villanueva et al. (2021)

Dragonfly: a rotorcraft for Titan

NASA's Dragonfly (launch 2028, arrival 2034) will explore Titan's surface.

- Nuclear-powered rotorcraft, the first of its kind for planetary exploration
- Mass: ~450 kg; can fly ~10 km between science stops
- Will land in the Selk impact crater region
- Sampling organic-rich dunes for mass spectrometry
- Investigating prebiotic chemistry: does the combination of complex organics and transient impact-melt water produce biomolecules?
- 2.7 year primary mission, visiting 20+ sites

Source: Barnes et al. (2021), *PSJ*

JWST: weather on distant worlds

JWST is transforming our view of exoplanet atmospheres.

- WASP-39 b: first clear detection of CO_2 in an exoplanet atmosphere (2022), then SO_2 from photochemistry (2023)
- WASP-107 b: SO_2 and silicate cloud signatures (2024)
- Phase curves map dayside-to-nightside heat redistribution on hot Jupiters
- Measurements of terminator cloudiness (east-west asymmetry)
- Searching for weather patterns, circulation signatures
- First steps toward characterising habitable-zone rocky planet atmospheres

Source: The JWST Transiting Exoplanet Community ERS Team (2023)

Recent advances: early Mars climate

- Updated 3D climate models with reducing gases (H_2, CH_4) can warm early Mars
- Collision-induced H_2-CO_2 absorption is a strong greenhouse at high CO_2 pressures (Wordsworth et al. 2017)
- Impact-triggered warming events: large impactors deliver heat and deflect vapour into the atmosphere
- Localised glaciation models: the “icy highlands” picture
- Episodic vs. sustained warming: still debated
- Perseverance (Jezero crater, 2021–) investigating the aqueous record

Source: Wordsworth (2016); Wordsworth et al. (2017)

Summary

1. Clouds form when rising air cools past the **saturation vapour pressure** at the LCL
2. The Clausius-Clapeyron equation gives $P_{\mathrm{sat}}(T) \propto \exp(-L_v/R_v T)$; governs all clouds
3. Cloud species vary: $\mathrm{H_2O}$ (Earth), $\mathrm{H_2SO_4}$ (Venus), $\mathrm{NH_3/H_2O}$ layers (Jupiter)
4. Atmospheric circulation: Hadley cells, Coriolis deflection, jet streams
5. Rotation rate sets number of circulation cells (1 on Venus, 3 on Earth, many on Jupiter)
6. Weather: Mars dust storms, Venus super-rotation, Jupiter GRS, Saturn hexagon
7. Faint young Sun paradox: early Sun 30% fainter, yet Earth had liquid water
8. Carbonate-silicate cycle: Earth's climate thermostat; failed on Venus and Mars

Looking ahead to Lecture 7

- Three lectures treated planets from the top down: energy budgets, vertical structure, and now clouds, winds, and climate.
- Lecture 7 turns to the solid surfaces beneath: impact craters as chronometers, volcanism and tectonics from within, erosion and weathering from above.
- The climate story returns as geology: Mars’s valley networks become a record to read, not only a puzzle to solve.

Questions?

Next: Lecture 7, Planetary surfaces: geology, geomorphology & geophysics

Mon 28 Sep, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

`https://ips.formingworlds.space`